Job Server User Guide

This is the user guide to use the MediciaSAFELY Job Server. At the moment, the server is running at http://rcdemo.medicia.ai:8000.

Steps to Use MediciaSAFELY

These are the steps the client should follow to use MediciaSAFELY and run jobs, here are the steps:

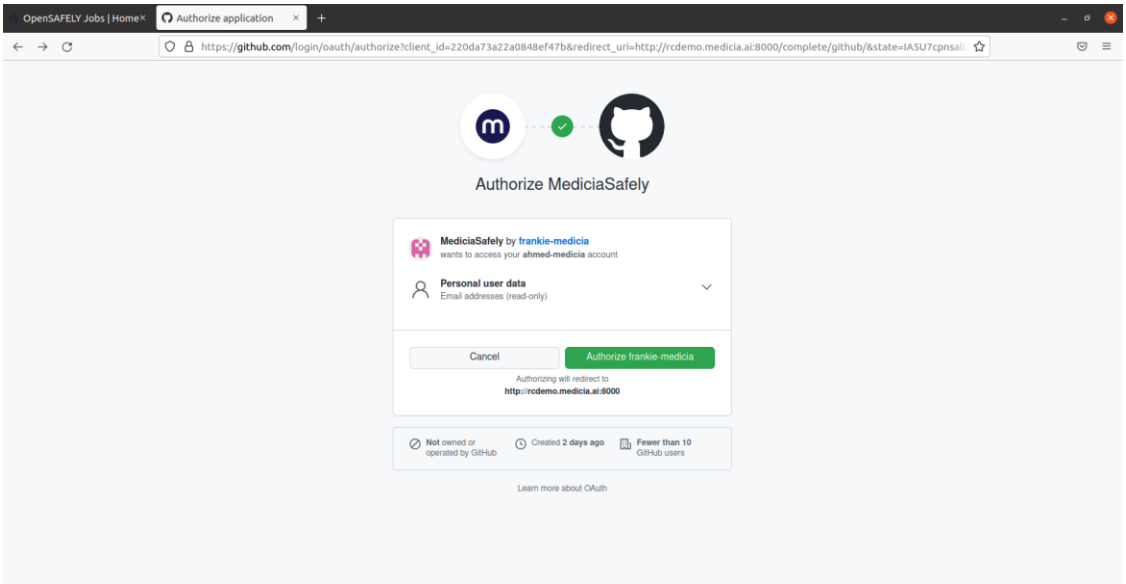
1. Client logins with GitHub
2. Client creates and submits an application/project.
3. Ask for a backend and organization.
4. Ask for adding project in the organization.
5. Ask for adding members to the project.
6. Edit project settings.
7. Create workspace.
8. Enter the GitHub PAT.
9. Run jobs.
10. View results.

The next sections covers these steps.

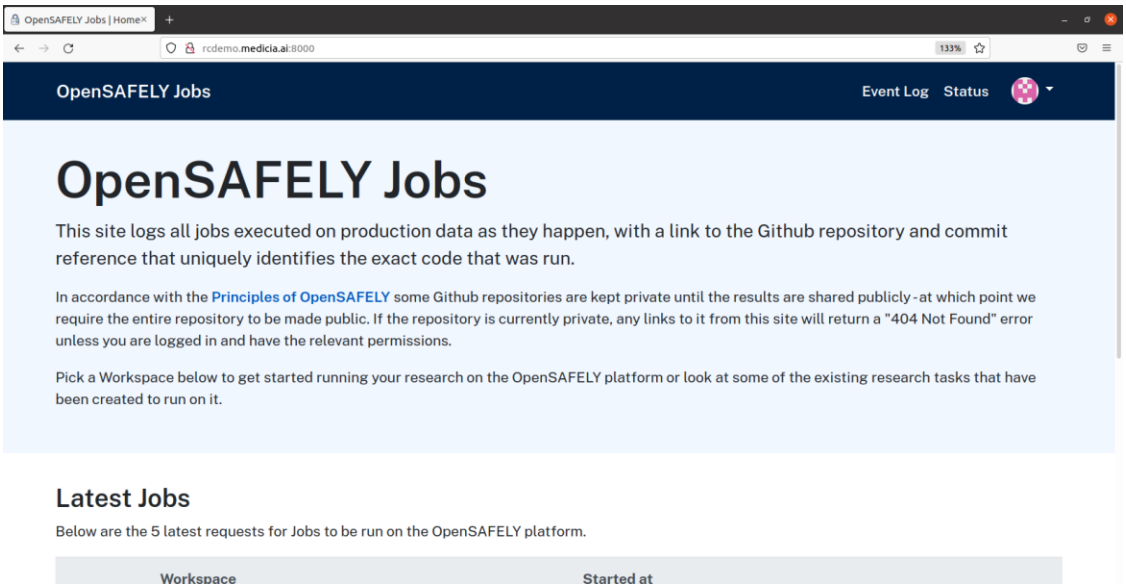
1. Client Login with GitHub

The client must login to MediciaSAFELY with their GitHub account:
http://rcdemo.medicia.ai:8000/login

Then go to the login page and click the login button. You will be directed to the GitHub login page. Then you will be asked for authorizing MediciaSAFELY to login with your GitHub account. Just click the green `Authorize ...` button.



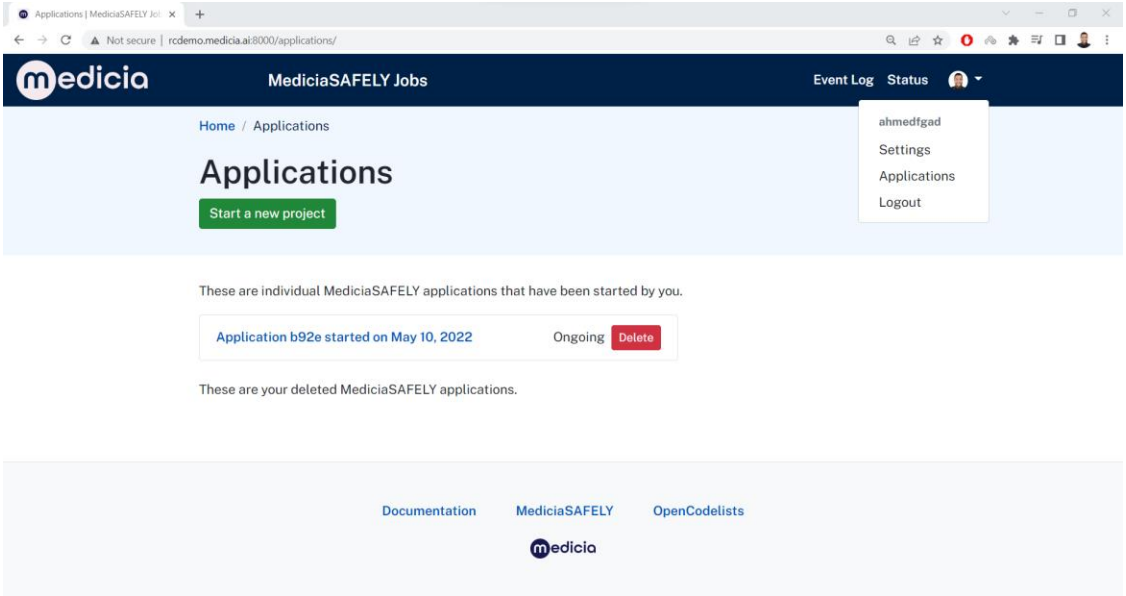
After authorizing the app, you will be redirected to the server with successful login. In the next image, the icon of the GitHub account appears at the top-right.



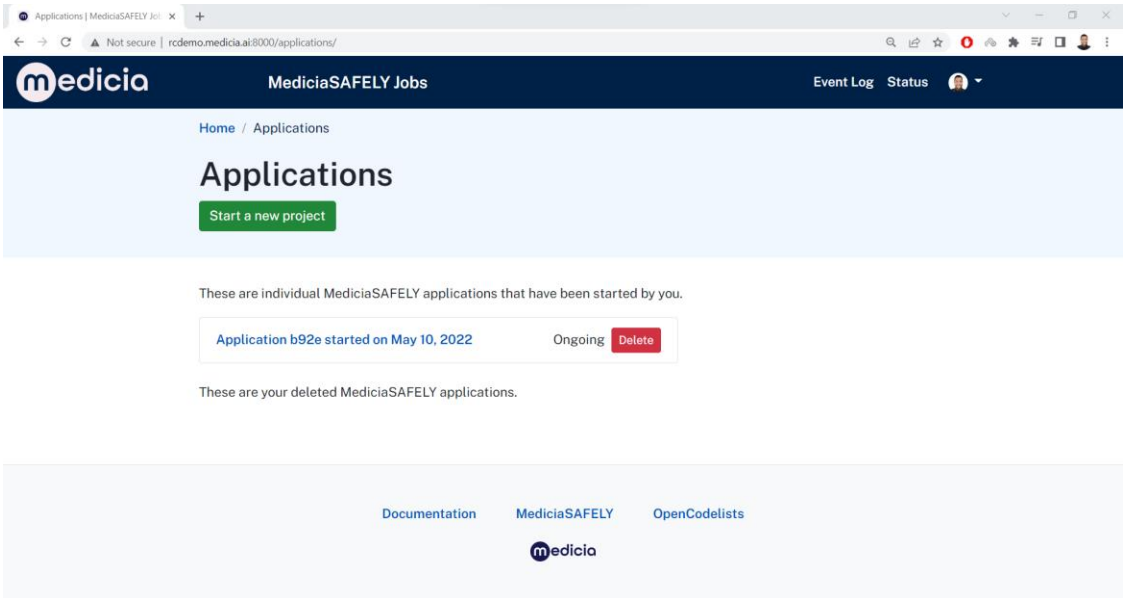
2. Client Creates and Submits an Application/Project

Once the client logins to MediciaSAFELY, then the next step is to create an application/Project. Just click the GitHub account icon at the top-right corner and a menu appears.

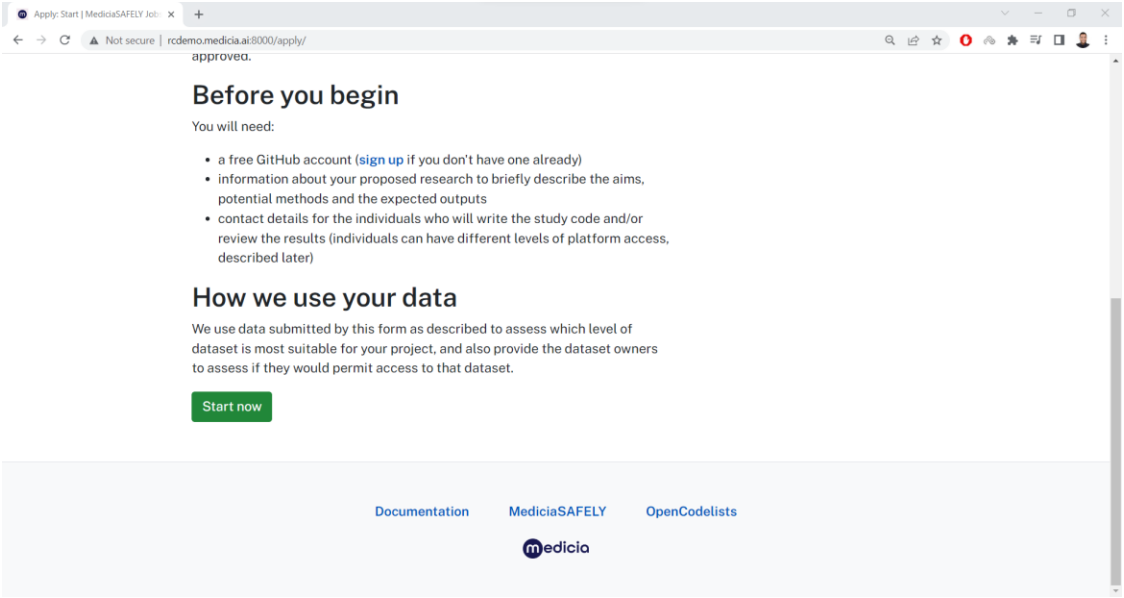
> If you are struggling creating an application, just contact the MediciaSAFELY team to create a project for you.



Click the `Applications` option which goes to this page: http://rcdemo.medicia.ai:8000/applications. THis page shows a list of applications that the user started.

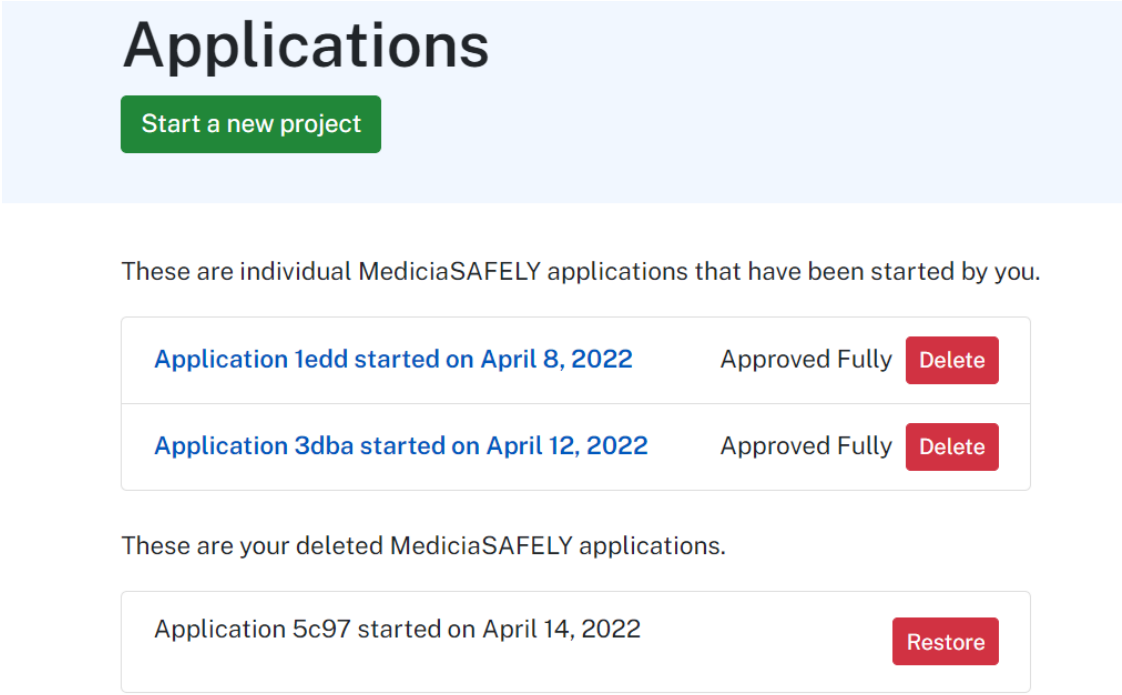


In the Applications page, click the `Start a new project` button which goes to http://rcdemo.medicia.ai:8000/apply. At the bottom of this page, click the `Start now` button to start creating an application.



Follow the steps until the application is submitted. Pay attention to the `Study name` value because it is the project name that the client will later use. This name what appears on the dashboard too.

Once the application is submitted, just wait for a MediciaSAFELY admin to approve it for you. The status of the applications appears in the http://rcdemo.medicia.ai:8000/applications page. When its status changes to `Approved Fully`, then the application is approved by an admin and ready to use. Now, you have a MediciaSAFELY project.



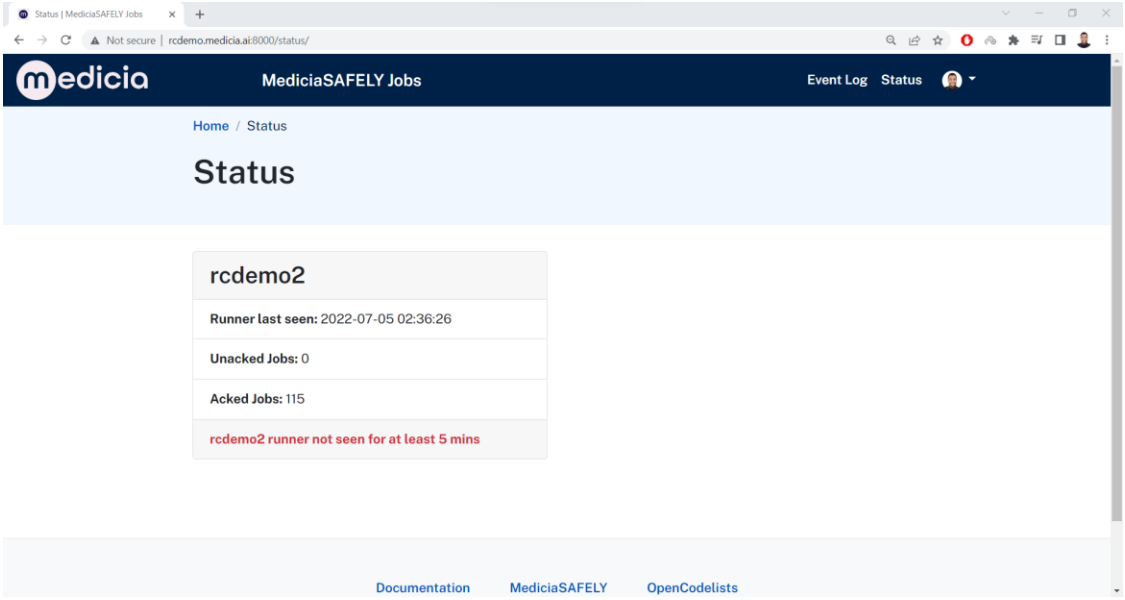
3. Ask for a Backend and Organization

Once the user owns a MediciaSAFELY project, next step is to ask for a MediciaSAFELY backend and organization. A MediciaSAFELY admin will take care of creating a backend and organization for you.

The admin is responsible for adding the users as a member in both the backend and organization.

The URL for the organization has this format `http://rcdemo.medicia.ai:8000/[org-name]` where `[org-name]` should be replaced by the organization name.

The user can check the backend he/she is a member by visiting this page: http://rcdemo.medicia.ai:8000/status. For example, the next image shows that the current user is a member of the `rcdemo2` backend.



4. Ask for Adding Project in the Organization

Before the user can use the project, a MediciaSAFELY admin must add the project in the organization. If not already added by an admin, contact the MediciaSAFELY team to ask for adding the project in your organization.

5. Ask for Adding Members to the Project

Only MediciaSAFELY admins can add members to the project. The regular user can only edit (remove or change permissions) users who are already members. But the user must be assigned the role/permission `Project Coordinator` to edit the members.

The user must ask a MediciaSAFELY admin to be added as a member the project.

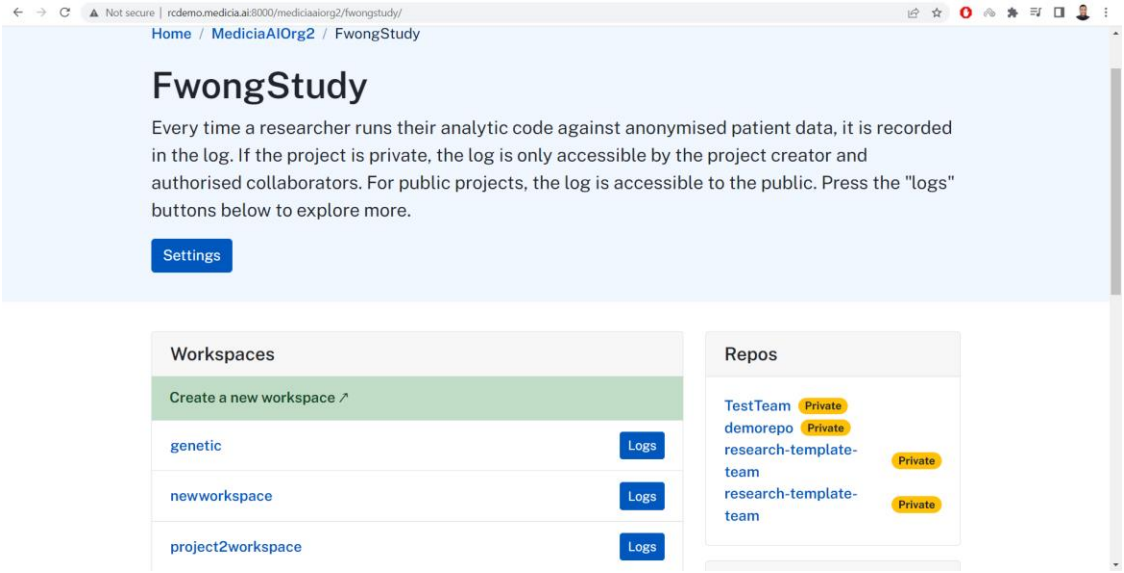
6. Edit Project Settings

For a regular user with the role `Project Coordinator` in the project, then this user can edit the project settings. This includes editing other members of the project by either:

1. Removing members from the project.
2. Changing the roles/permissions of the members.

To do that, the first step is to go to the organization page. In the organization page, there will be a list of projects in this organization. For example, the next image shows that the `MediciaAIOrg2` organization has 3 projects.

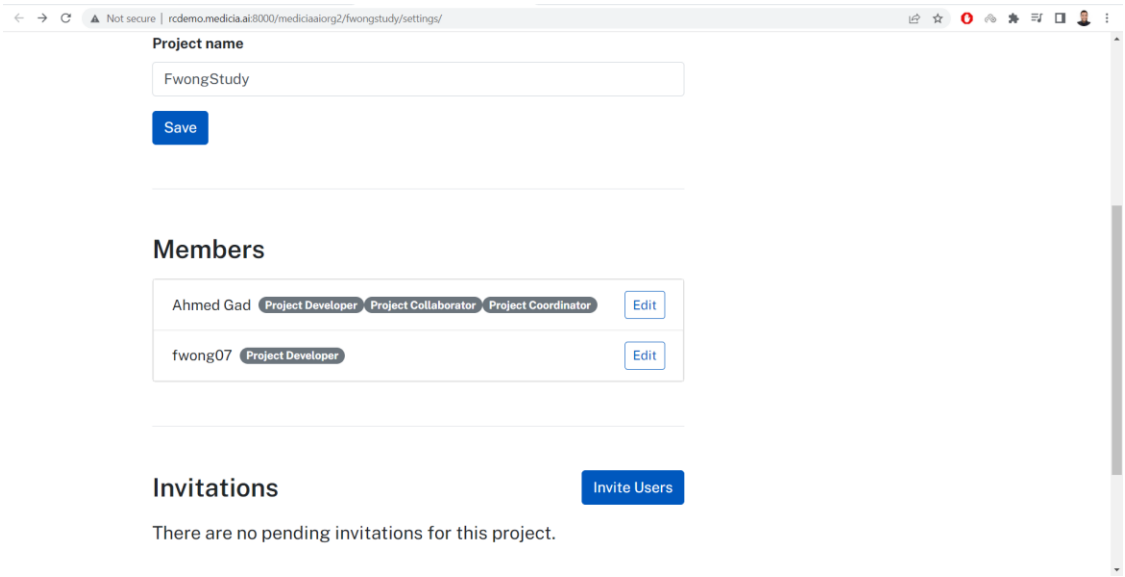
For example, the current user is a member only in the `FwongStudy` project. After clicking this project, it goes to the project's page: http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy



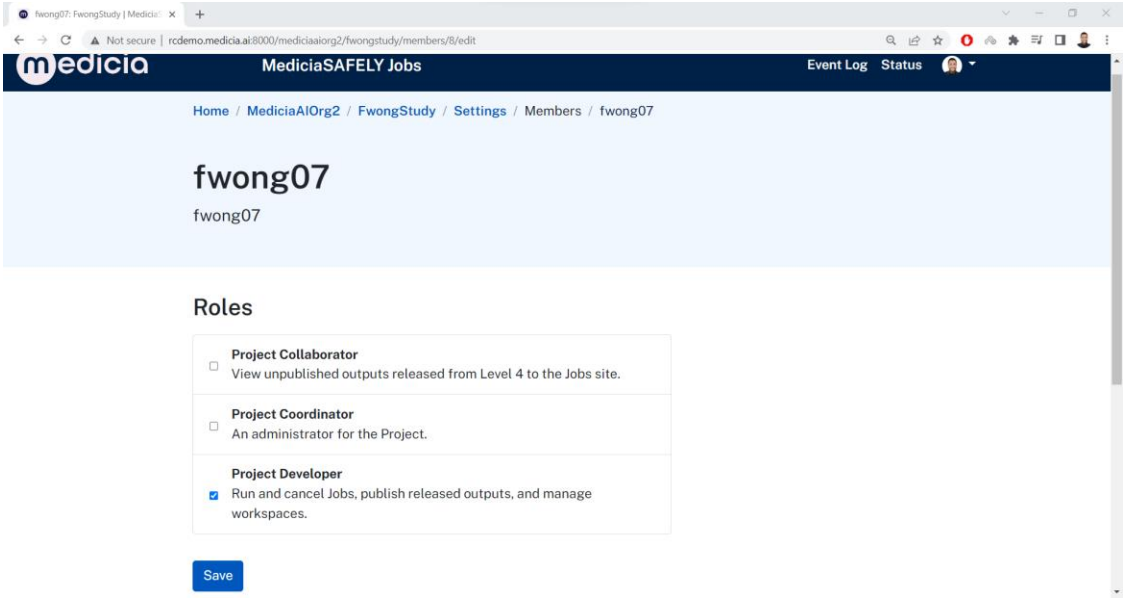
Given that the current user has the `Project Coordinator` role, then this user can edit the members. Just click the `Settings` button which goes to this page (http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/settings) as in the next image. There the user can change the project's settings including:

1. Project name

2. Editing the members by adding or removing members.
3. Change the role of the members of the project.



Let's click the `Edit` button besides the `fwong07` user. This goes to the user's page under the `FwongStudy` project. Select which roles to be assigned to the user and click the `Save` button.



In the project's settings page, the user can control 3 roles:

1. Project Collaborator.
2. Project Coordinator.
3. project Developer.

These 3 roles only applies to the current project. It may happen that the user is a member of more than 1 project. If the user is assigned a role as described above, then this role is only applied to the current project. But the user will not have this role in the other projects.

To assign the user roles that apply to all project, then contact a MediciaSAFELY admin to assign the user the roles that get applied to all projects.

This is a list of all roles (10) that can be assigned to the users.

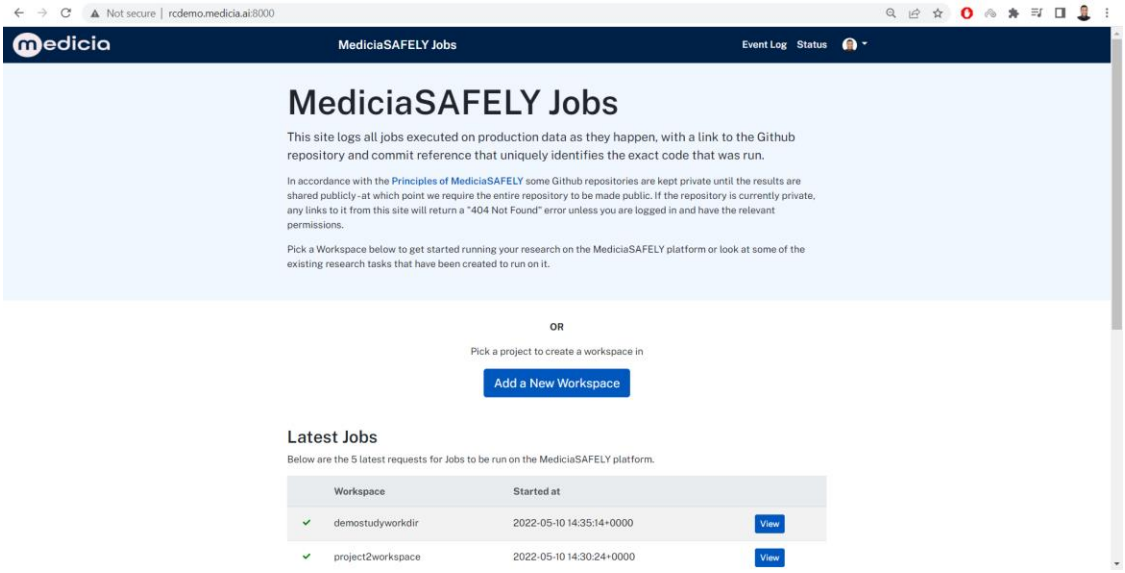
1. Core Developer.
2. Data Investigator.
3. Governance Reviewer.
4. Onboarding Agent.
5. Output Checker.
6. Output Publisher.
7. Project Collaborator.
8. Project Coordinator.
9. Project Developer.
10. Technical Reviewer.

As a summary, there are 3 roles can only work to either individual or all projects. The other roles can work only across all current and future projects.

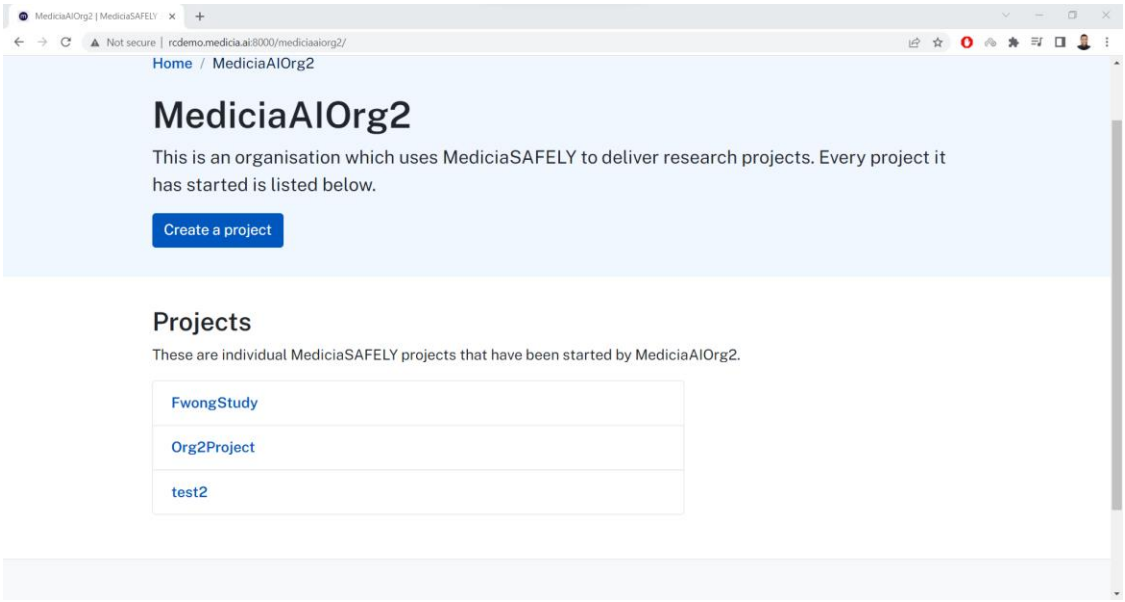
7. Create Workspace

The regular user can now login with their GitHub account and visit the homepage of MediciaSAFELY: http://rcdemo.medicia.ai:8000

Scroll down and click the `Add a New Workspace` button.

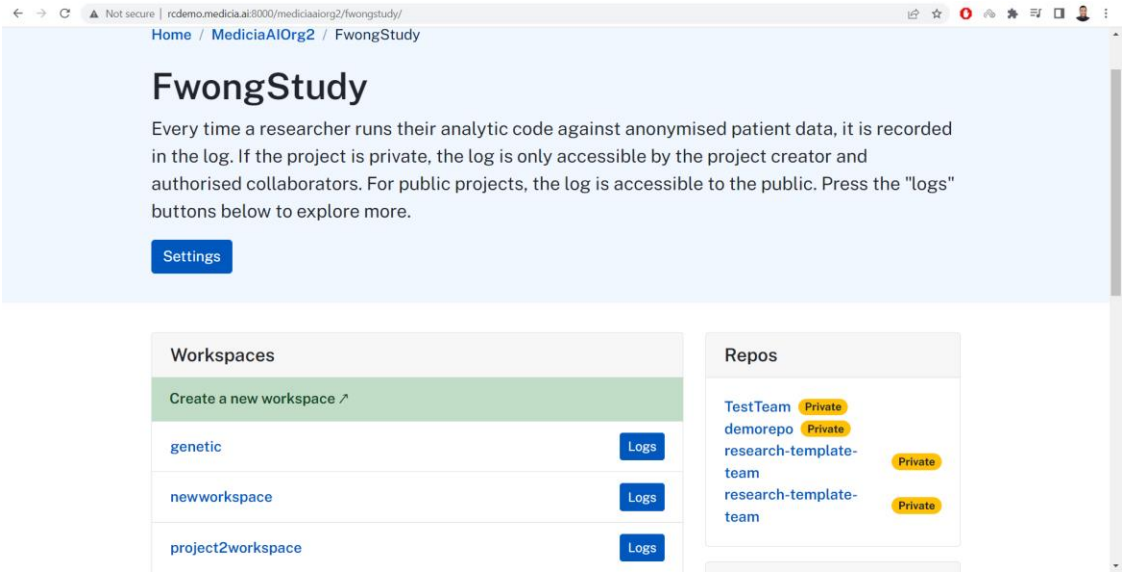


Clicking this button forwards the user to the its organization. For example, if the user is a member of the `mediciaaiorg2` organization, then it will be forwarded to this page: http://rcdemo.medicia.ai:8000/mediciaaiorg2



Click on the target project. For example, `FwongStudy` http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy. This forwards to the project's details page where all project's workspaces are listed.

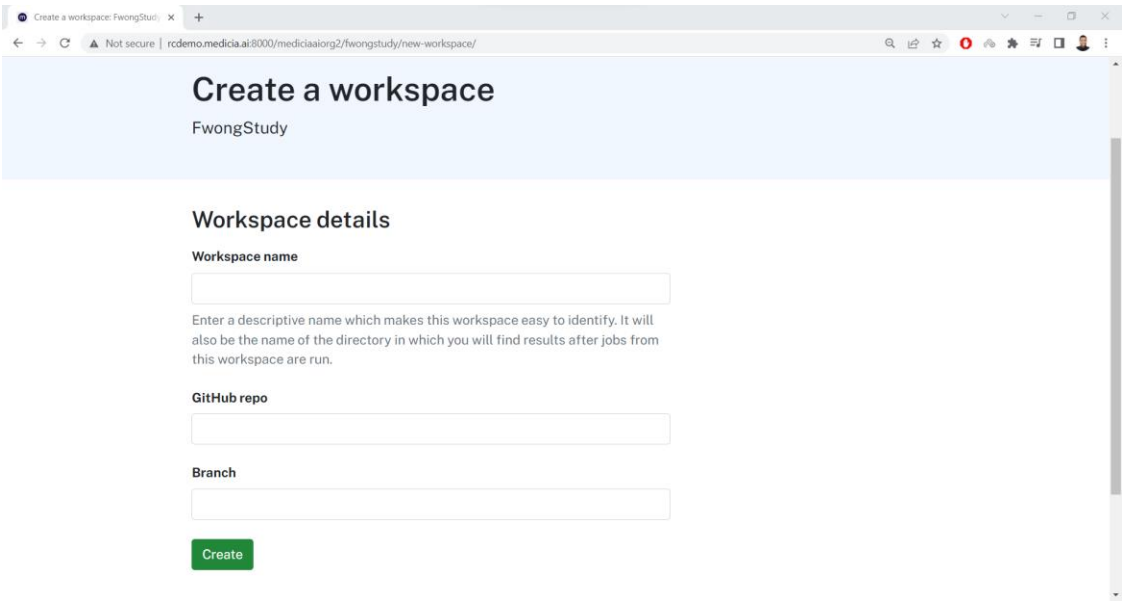
If the user is authorized to create workspaces, then it will find the `Create a new workspace ↗`. button there. Click on it: http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/new-workspace/



The user will be forwarded to a page to create a new workspace. Enter the following information into the workspace form:

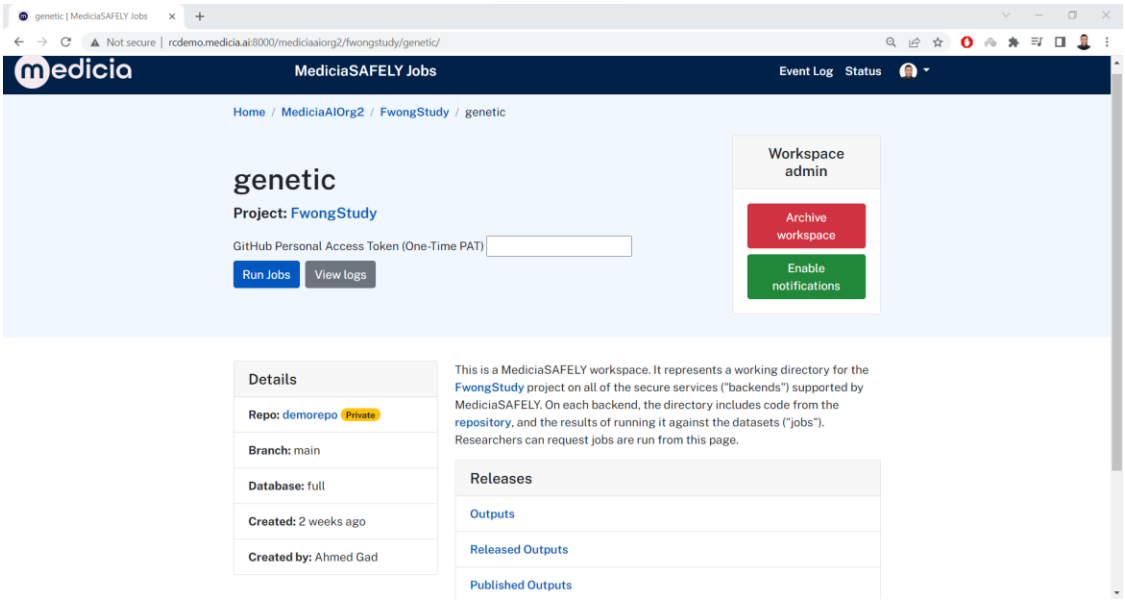
1. Workspace name. This is a name of your choice.
2. GitHub repository. This is the full URL of the GitHub repository. For example, https://github.com/MediciaAI/TestTeam.
3. Branch of the repository. For example, `main`.

Then click the `Create` button.

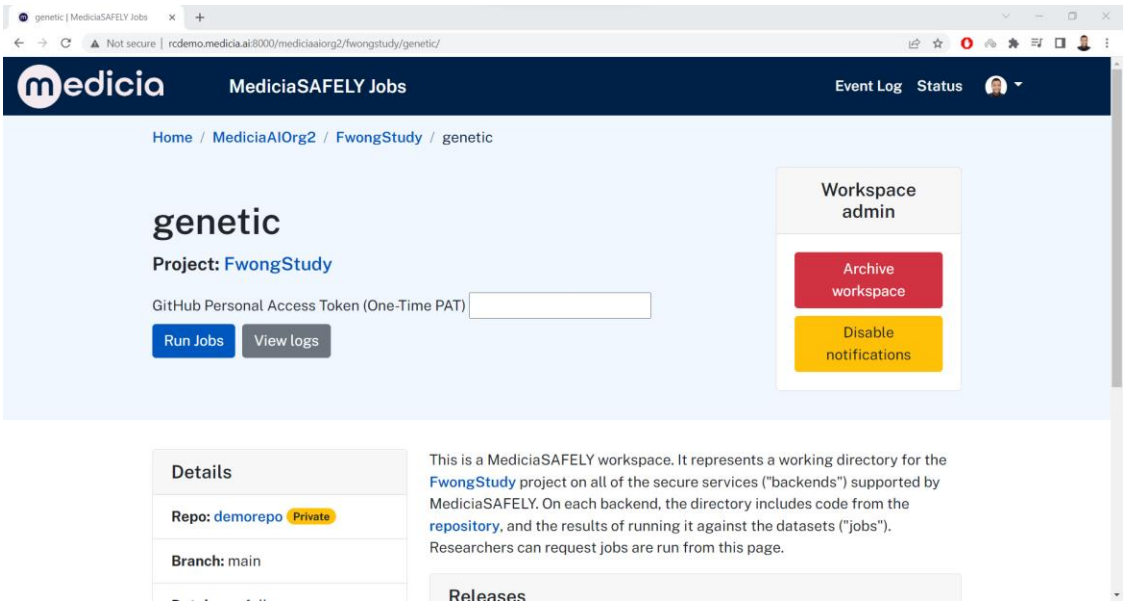


After the workspace is created, then go back to the project page where you will find the new workspace listed: http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy

Then click the created workspace:
http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/project2workspace



Email notifications are not enabled by default. To get notifications about all the jobs in this workspace, click on the `Enable notifications` button. Now, the button text becomes `Disable notifications`.

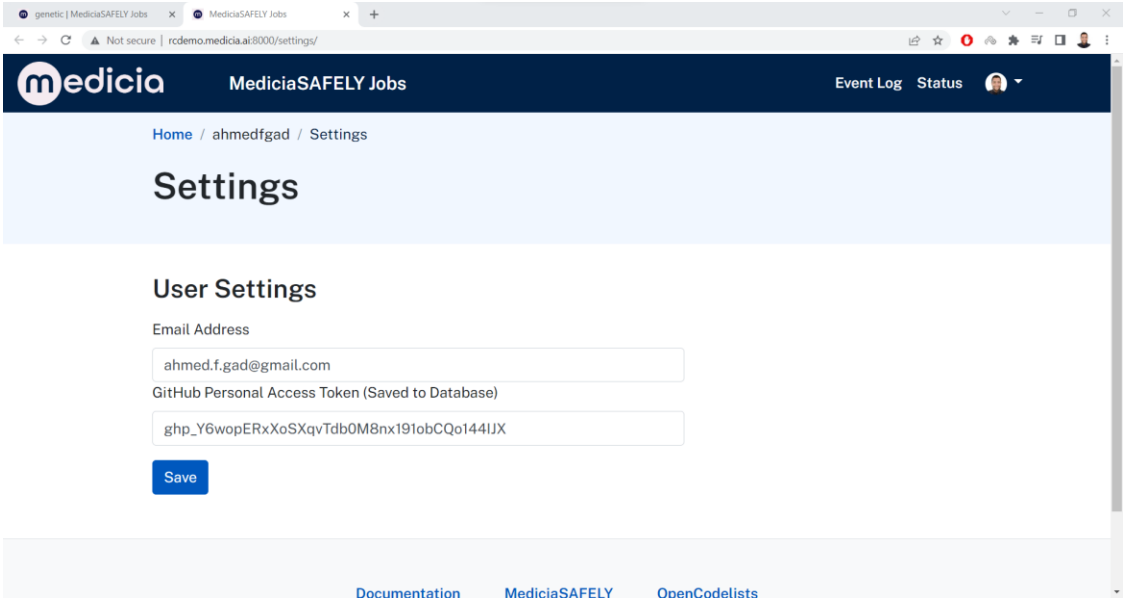


To archive the workspace, click the `Archive workspace` button.

8. Enter the GitHub PAT

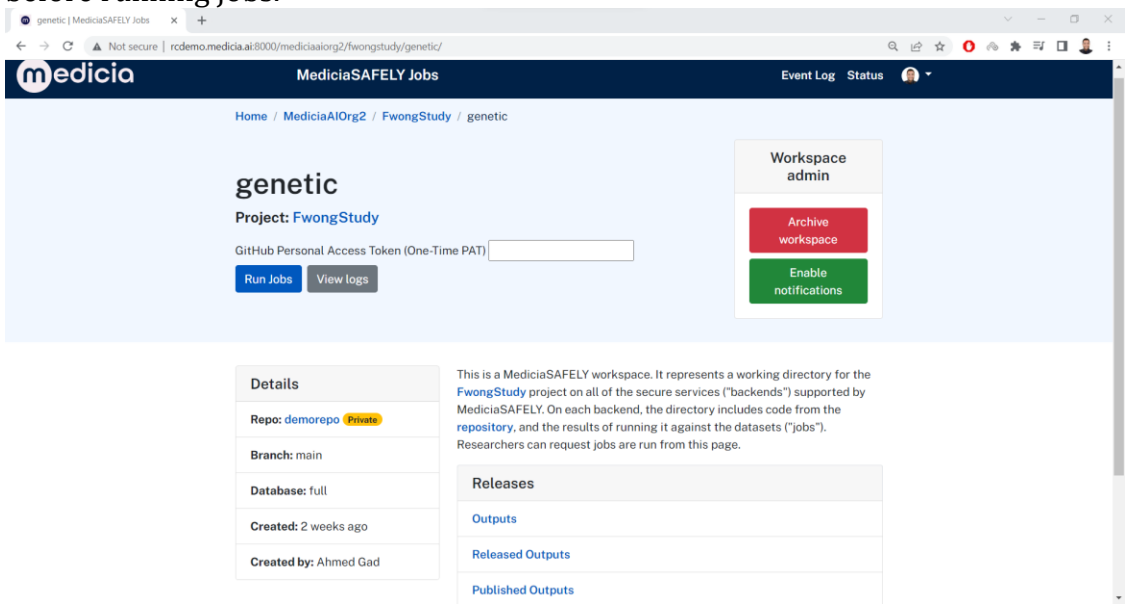
To grant the server access to the GitHub repository, you must provide the GitHub personal access token (PAT). There are 2 ways to give the PAT to the server:

1. Through the user's settings page. There is an input field to accept the PAT. After clicking the `Save` button, the PAT will be saved permanently to the database. In this case, the user enters the PAT only once and use it many times.
   The user can also change the email address to which notifications are sent.



2. If the user prefers not to keep the PAT stored in the database, then it can enter it in the workspace details page. There is an input field that accepts the PAT. This PAT is used only once and not saved in the database. But the user have to enter the PAT

before running jobs.

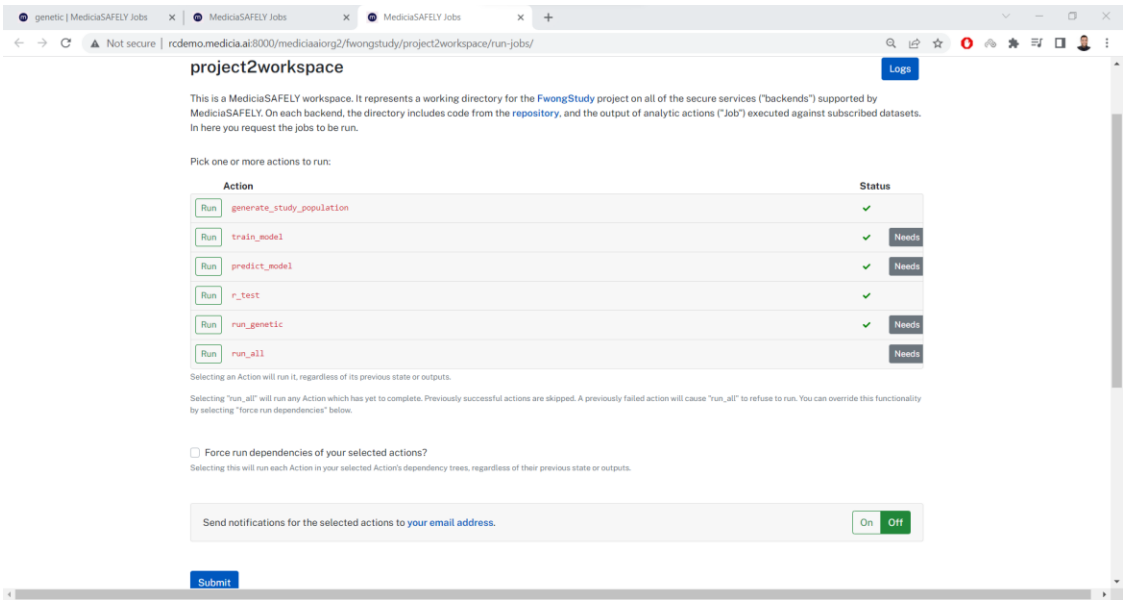


As a summary, if the PAT is not saved in the database, then the user have to enter it each time a job is to be run.

9. Run Jobs

Once the PAT is given, then the user can click the `Run Jobs` button in the workspace page. For example, http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/project2workspace.

This opens this page (http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/project2workspace/run-jobs/) which lists all the actions in the project according to the `project.yaml` file in the GitHub repository.



Click the `Run` button at the left side of the actions you would like to run. These buttons should be turned into green.

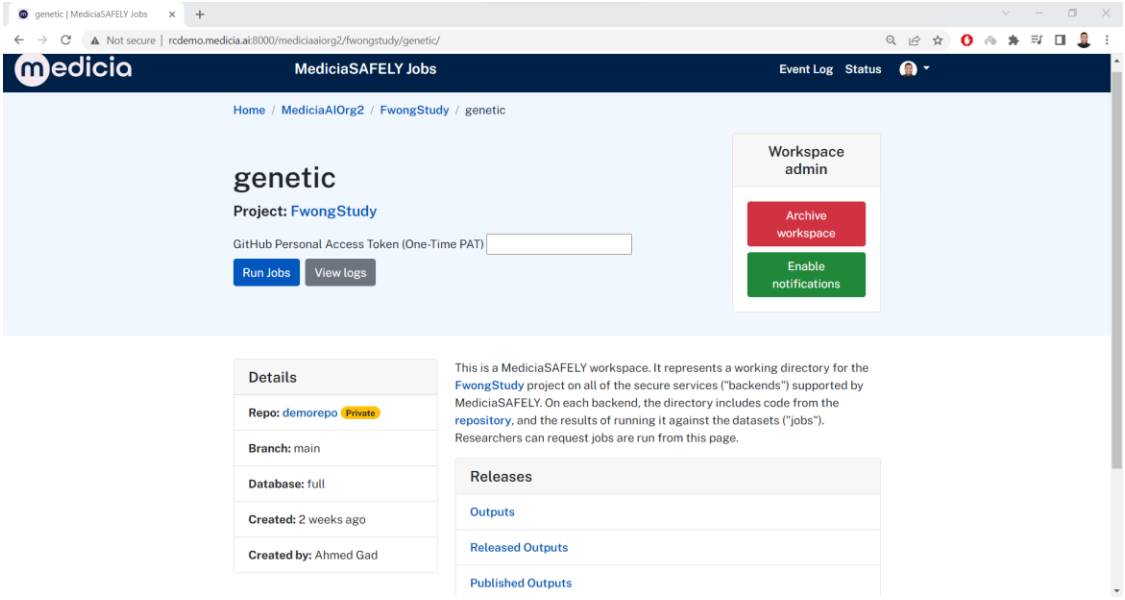
Check whether notifications should be sent to the email once each job is completed and change the state of the checkbox.

Then click `Submit`. Now, the jobs will be submitted from the job server to the job runner to be executed.

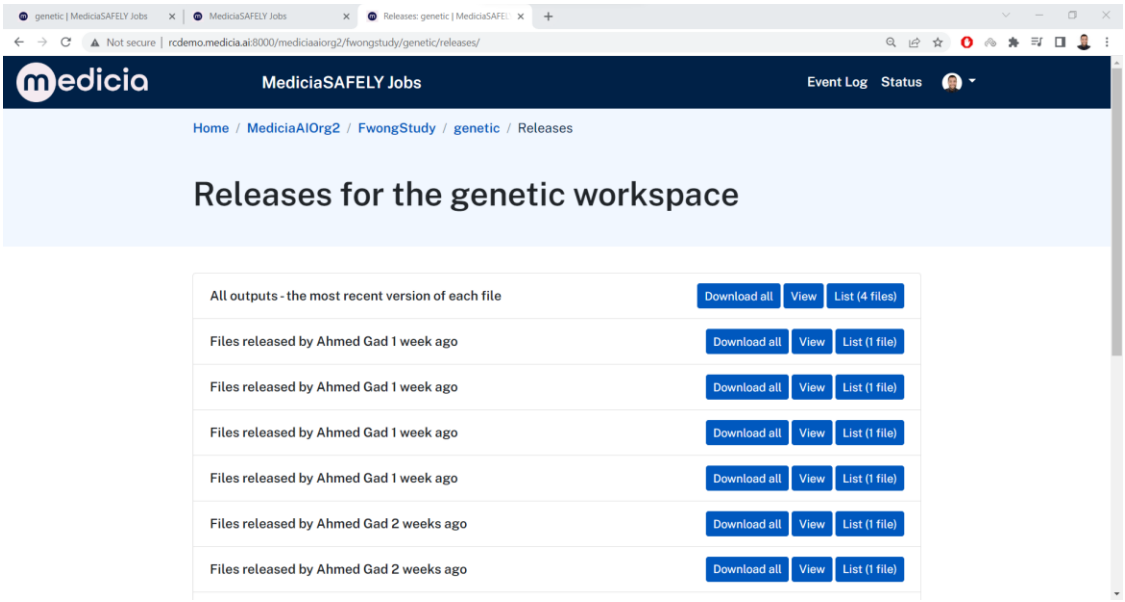
To check the status of the submitted jobs, go to the workspace logs page. For example, http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/project2workspace/logs/?next=/gad. A job is successfully completed if a green mark is shown.

10. View Results

Once the jobs are completed successfully, go back to the workspace page. Under the `Releases` section, click the `Released Outputs` page. There you will find the list of released files out of the completed jobs.

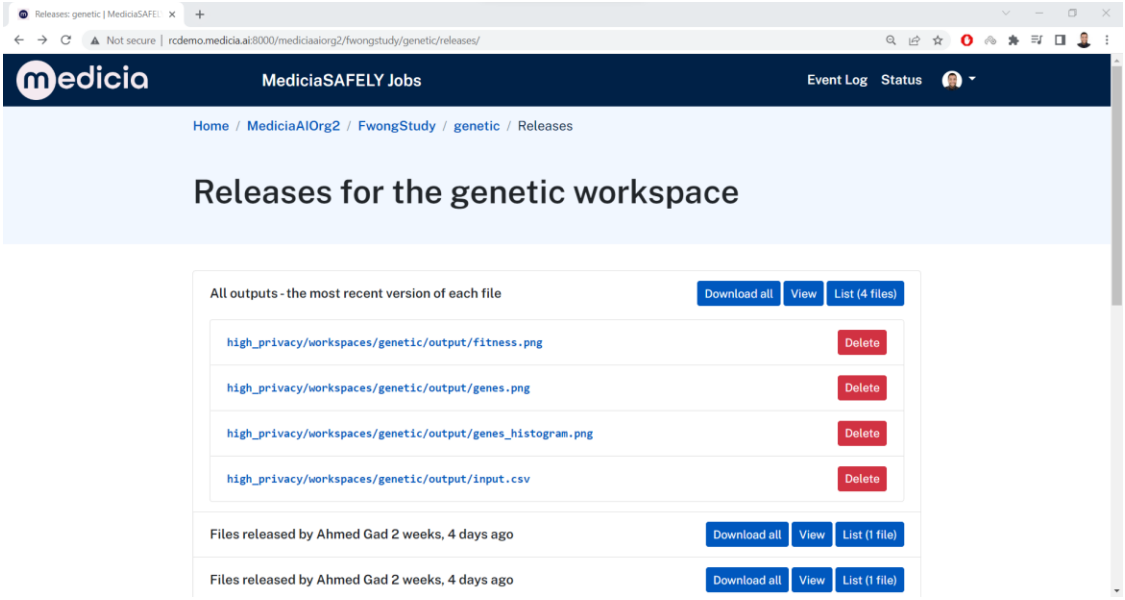


For example, this is the list of outputs in the `genetic` workspace: http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/genetic/releases. You can view the released outputs online or download them.

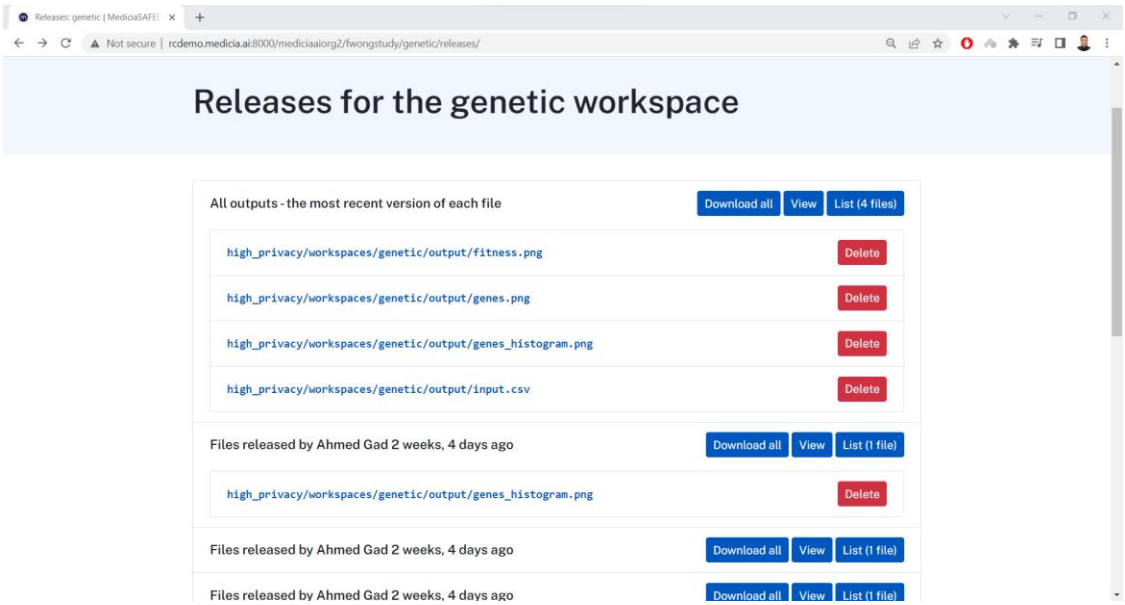


List the Released Files

The first row in the workspace releases has the most recent versions of all the released files in the workspace. Clicking on the `List` button lists all the released files.



By clicking the `List` button in the other rows, it only shows the individual file in this release.

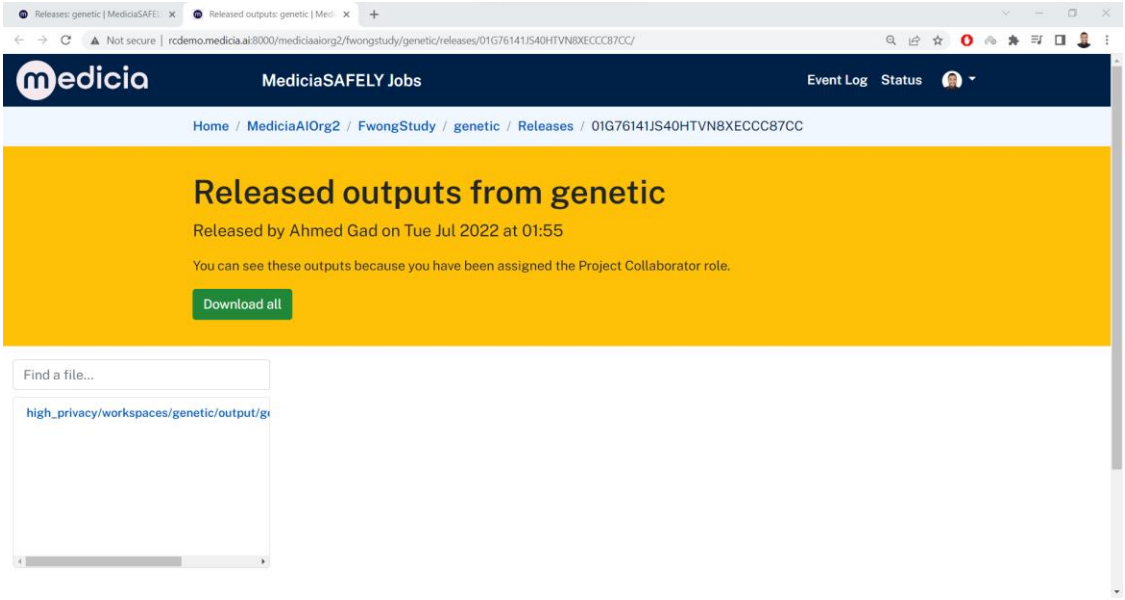


Click the `Delete` button besides each file to delete it from the workspace.

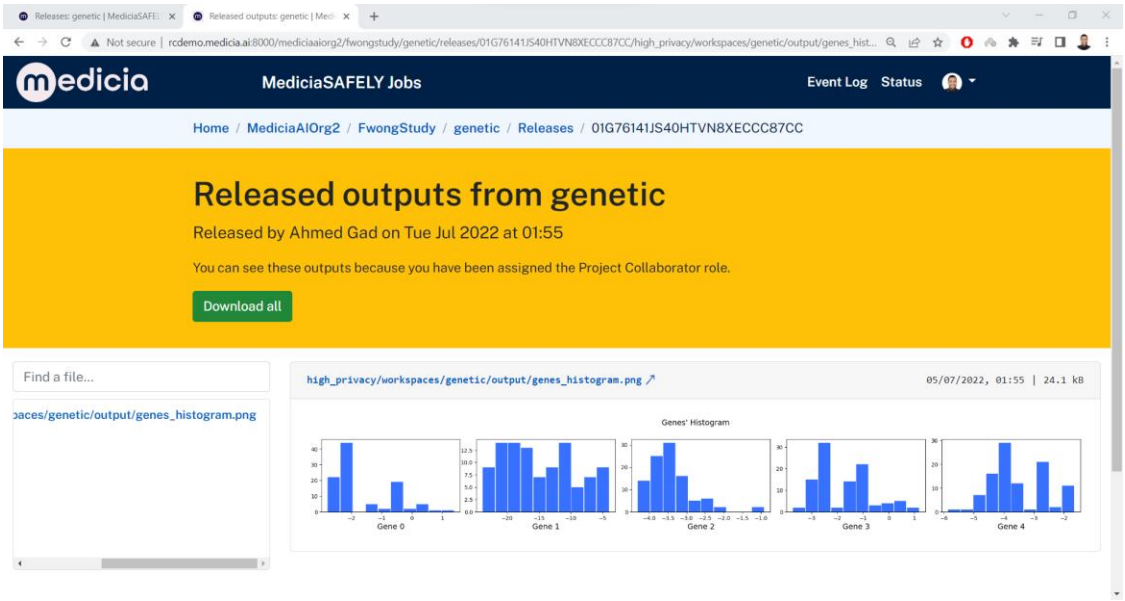
View the Released Files

Click on the `View` button besides each release to go to the release page. This is an example. The URL for this release is http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/genetic/releases/01G76141JS

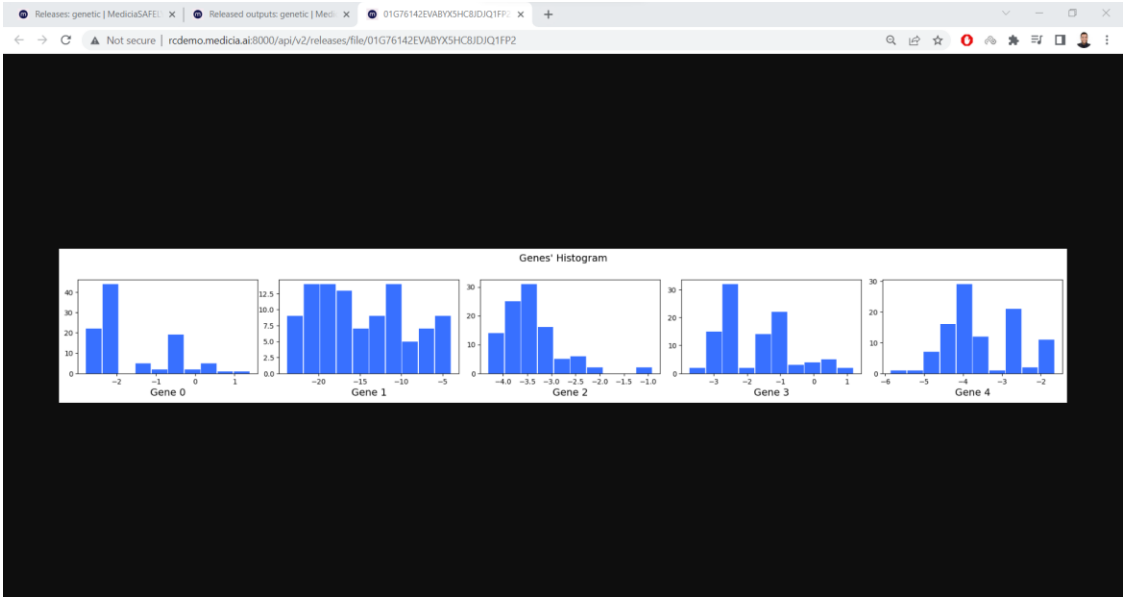
40HTVN8XECCC87CC. The code at the end of the URL `01G76141JS40HTVN8XECCC87CC` represents the release ID.



At the bottom-left side, there will be a list of files in this release. This release has a single file named `high_privacy/workspaces/genetic/output/genes_histogram.png`. Its URL is http://rcdemo.medicia.ai:8000/api/v2/releases/file/01G76142EVABYX5HC8JDJQ1FP2. The code at the end of the URL `01G76142EVABYX5HC8JDJQ1FP2` represents the file ID. Click on the file to view it at the bottom-right area.

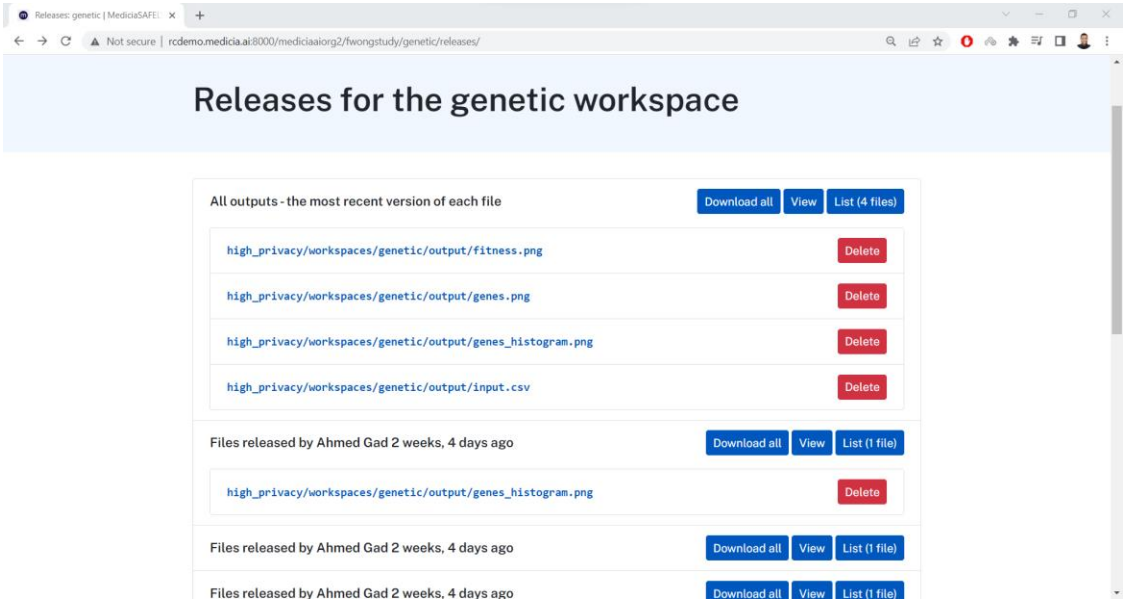


To view the file in a new tab, just click the link above it.

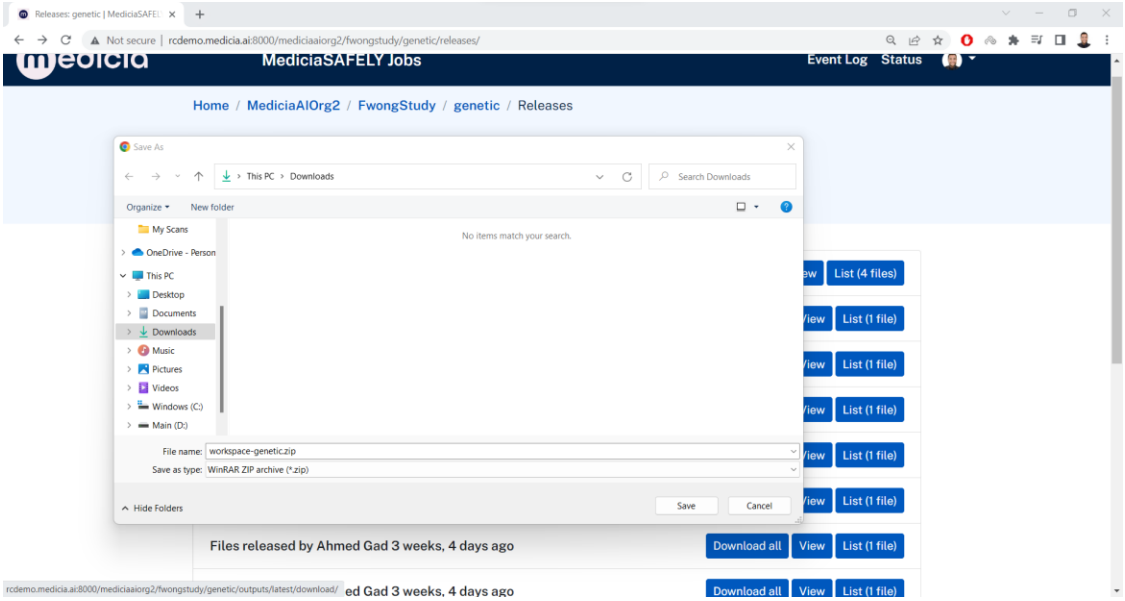


Download the Released Files

In the workspace releases page, click the `Download all` button to download all the files in a release.

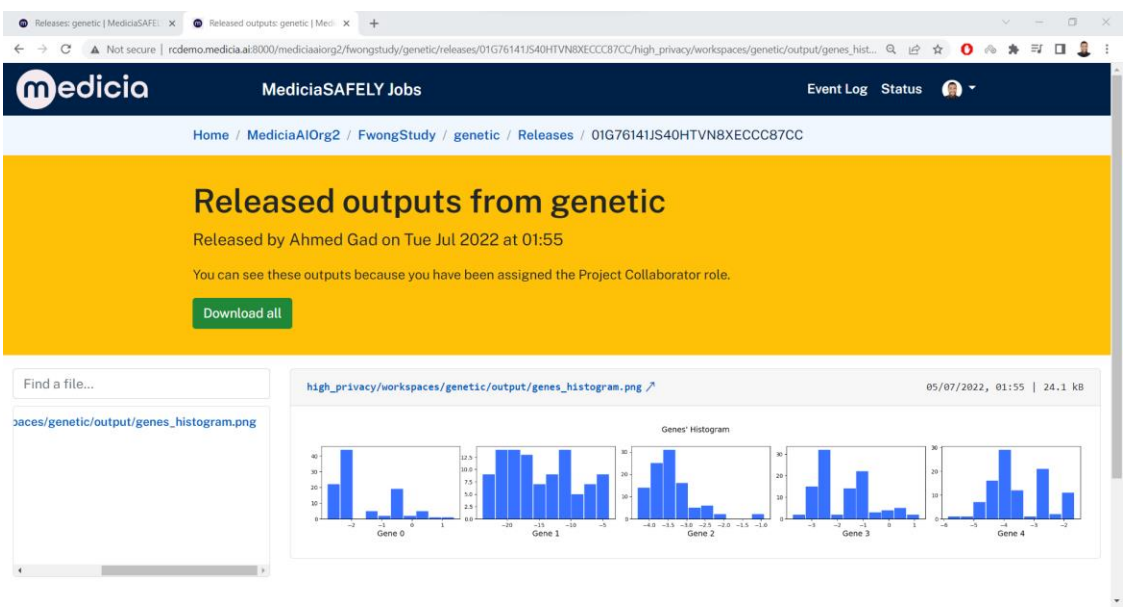


After clicking the `Download all` button, then all the files in the release will be added to a compressed file and you will be asked to choose the location to save it.



Once downloaded, just decompress the file. Follow the chain of folders until you open the `output` folder where the files exist. Then open the files locally on your machine.

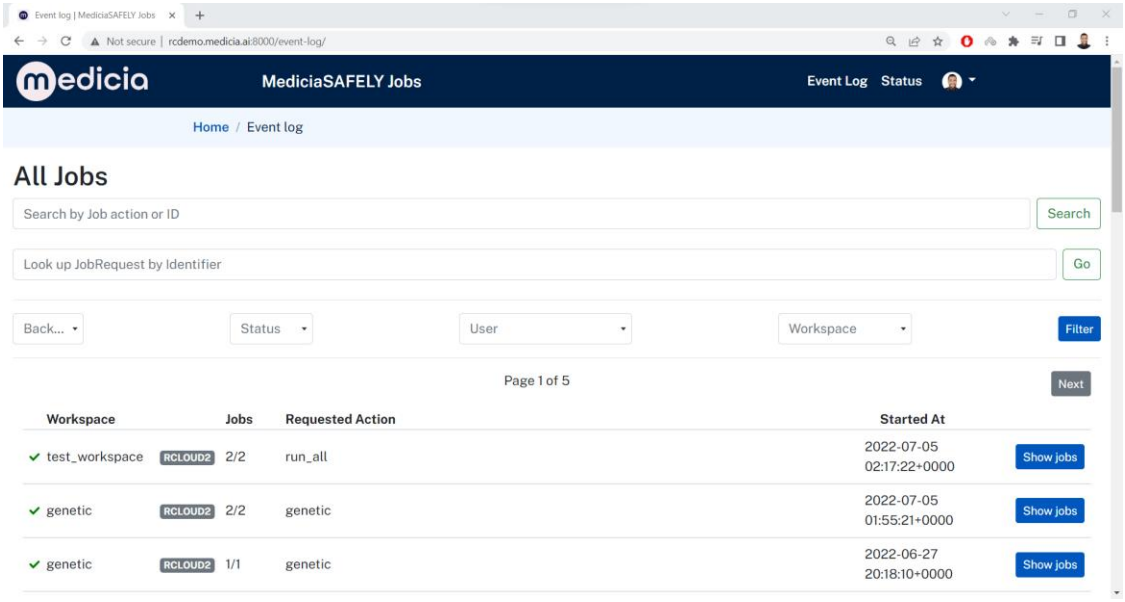
We can also download a specific file by clicking on the `View` button. There will be a `Download all` button to download the files in this release.



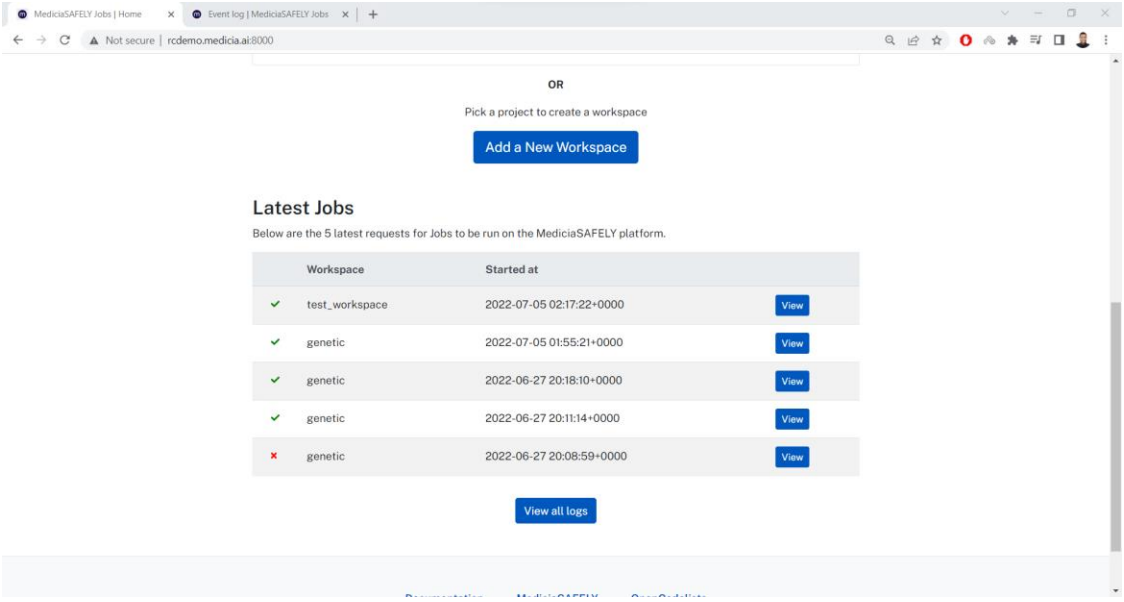
Event Log

At the top-right corner, click the `Event Log` link to view the logs. This page shows the logs of all the jobs across all workspaces. Each row in the table shows the following information:

1. The workspace name.
2. The backend name.
3. The number of jobs to be executed.
4. The requested action selected by the user. The actions are defined in the `project.yaml` file of the research study.
5. The time at which the job started.



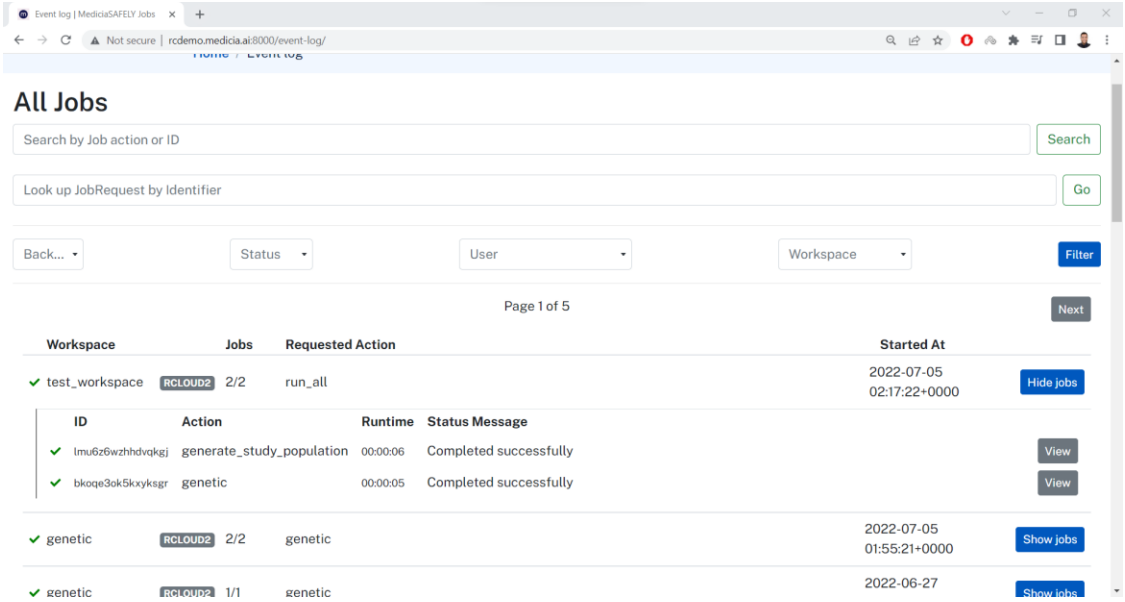
We can access this page also from the the homepage of the server (http://rcdemo.medicia.ai:8000). As in the next image, there is a list of the latest 5 job requests across all workspaces of the user. Below the list, there is a button named `View all logs`.



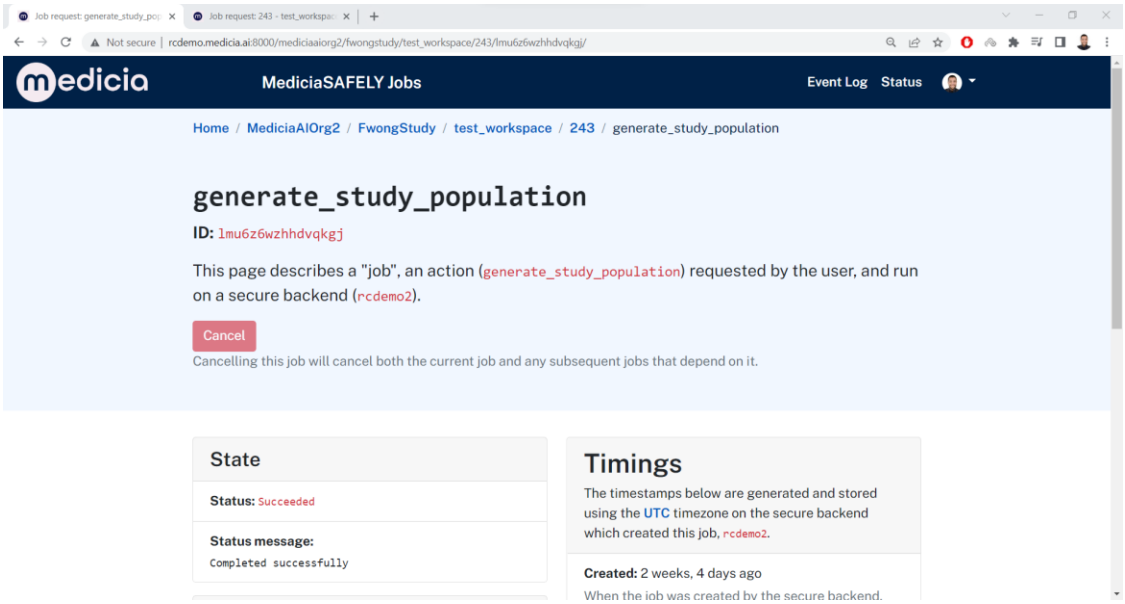
Clicking on this link goes to the events log page (http://rcdemo.medicia.ai:8000/event-log).

Click on the `Show jobs` button at the right of each row to show the jobs handled. For example, there are 2 jobs to be executed in the first row. The new table shows the following information:

1. Job request ID.
2. Action name.
3. Time consumed to complete the action.
4. A message indicating the status of the action.



Also click the `View` button in gray to see more information about the job request. For example, click the `View` button of the job request with ID `lmu6z6wzhhdvqkgj` goes to this page: http://rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/test_workspace/243/lmu6z6wzhhdvqkgj.

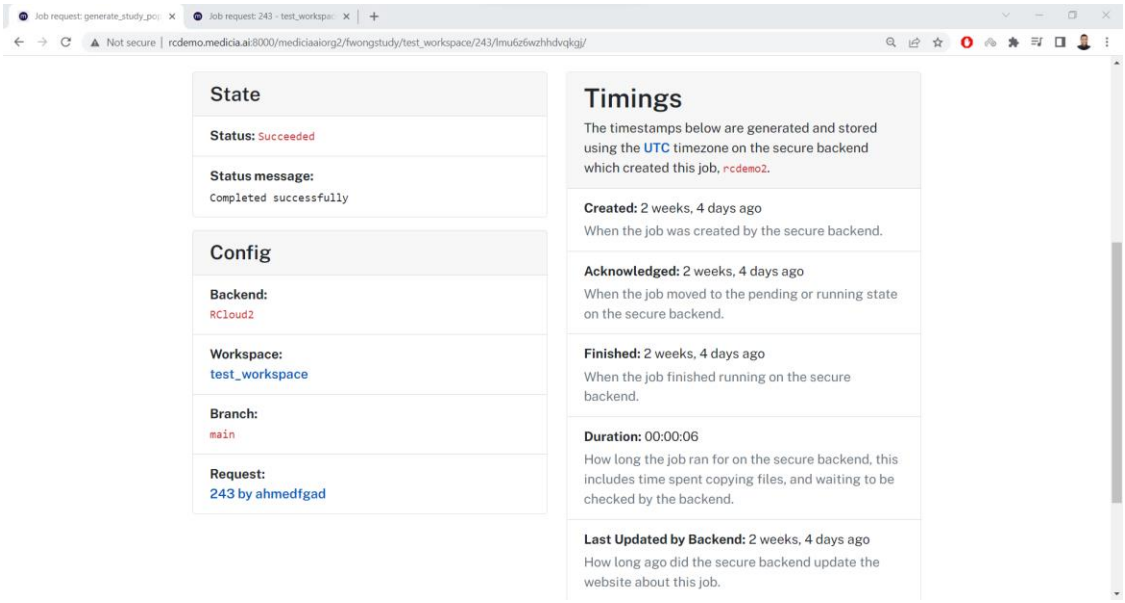


This page has 3 main sections:

1. State: Shows the state of the job.

2. Config: Configuration of the job.

3. Timings.



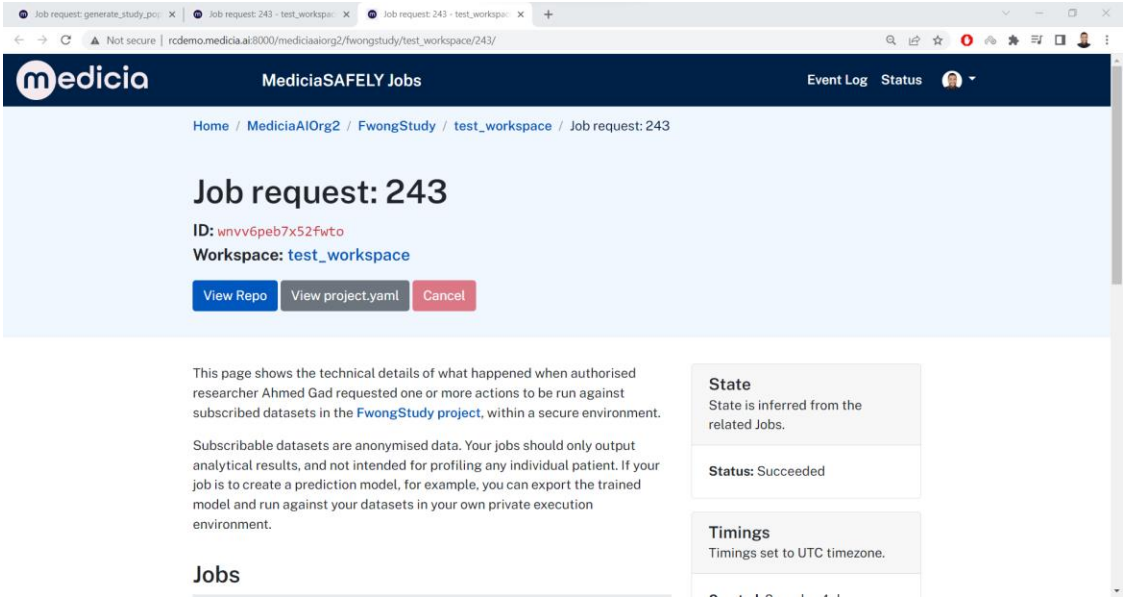
The last entry in the `Config` section (243 by ahmedfgad) is a link to the job request. This is the page of the job request with ID `243`.

At the top, there are 3 buttons:

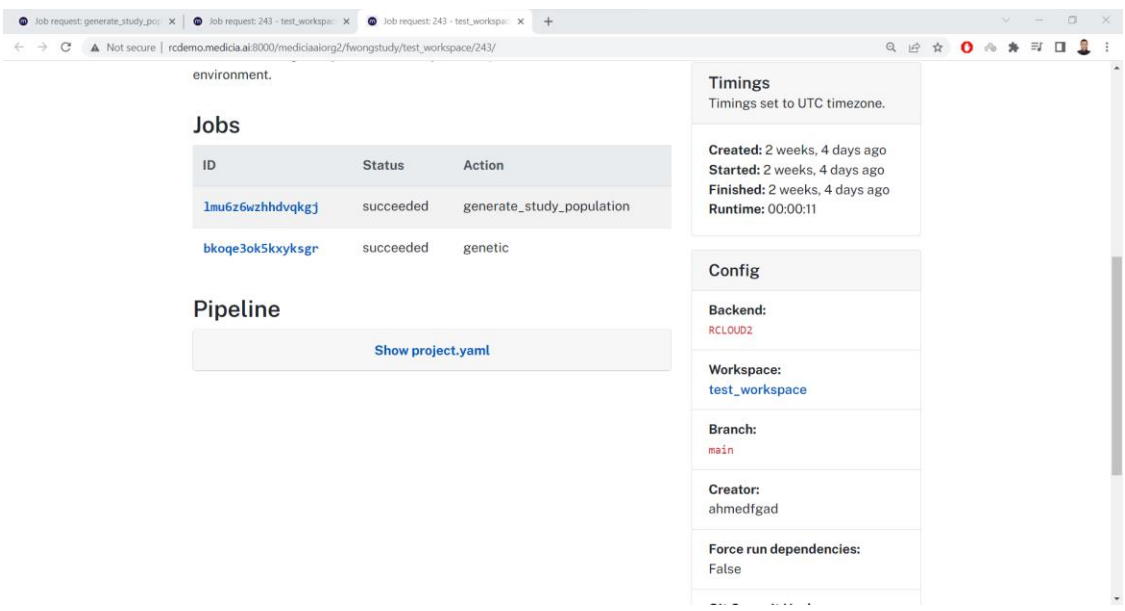
1. `View Repo`: This is a link to the GitHub repository.

2. `View project.yaml`: A link to view the `project.yaml` file in the GitHub repository.

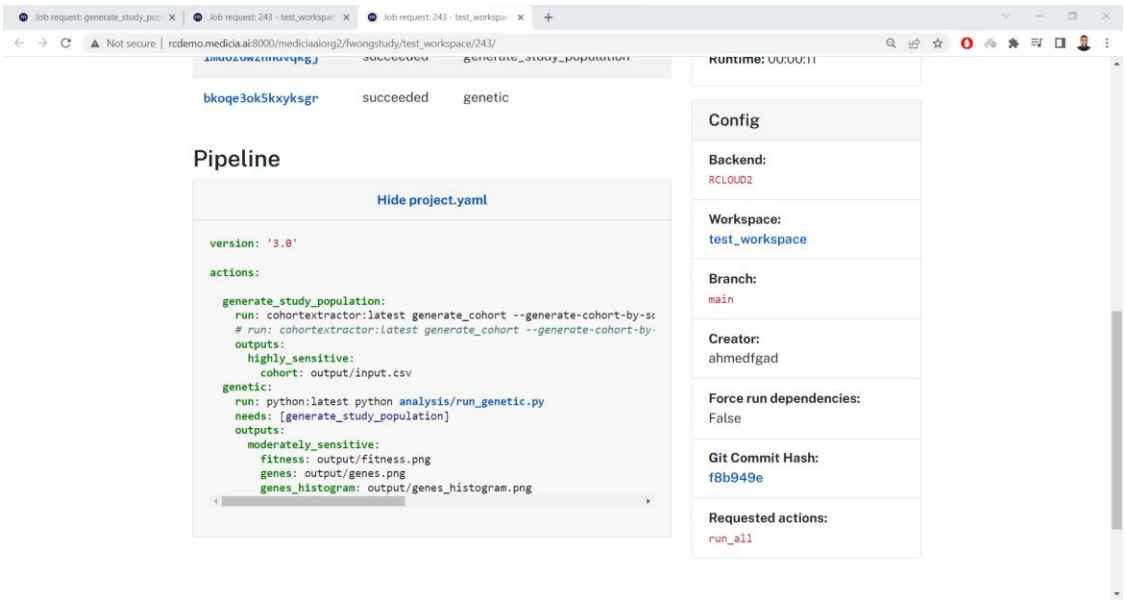
3. `Cancel`: If the job request is still in progress, click on this button to cancel it. This button is disabled because the job request is already handled.



Down in the page, there is a list of the jobs in this job request. There is also a button called `Show project.yaml` to view the `project.yaml` file.



Just click this button to view the `project.yaml` file in the page.

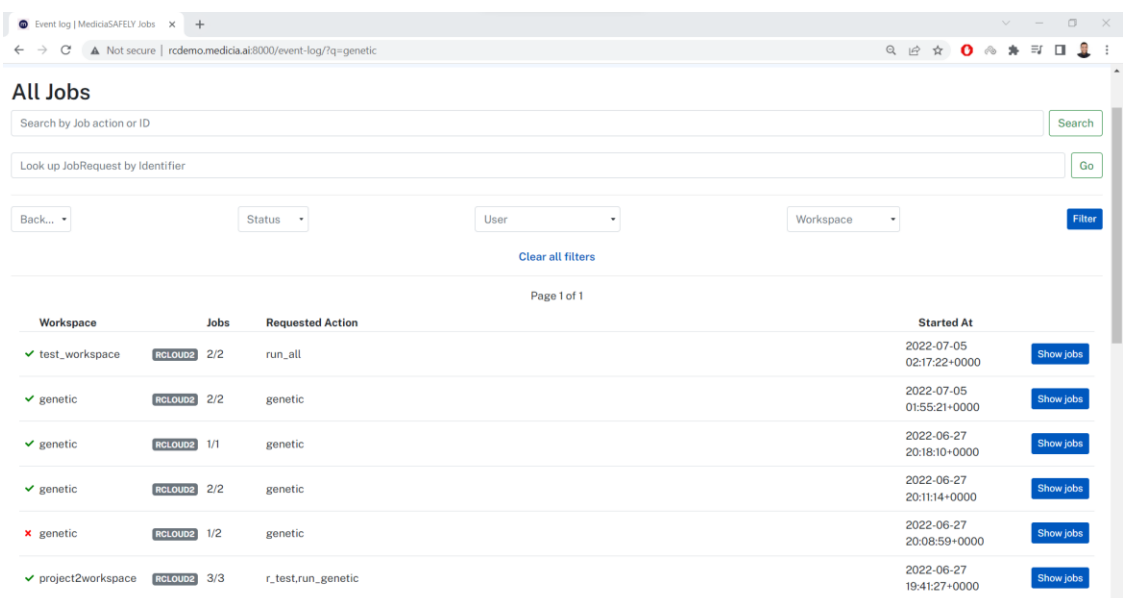


Filter the Logs

There are 2 inputs fields in the events log page:

1. The first field accepts the action name or job ID.
2. The second fields accepts the job request ID.

Just enter the value and click on the button at the right side of the input field. For example, the next image shows the result of filtering the jobs whose action is `genetic`.

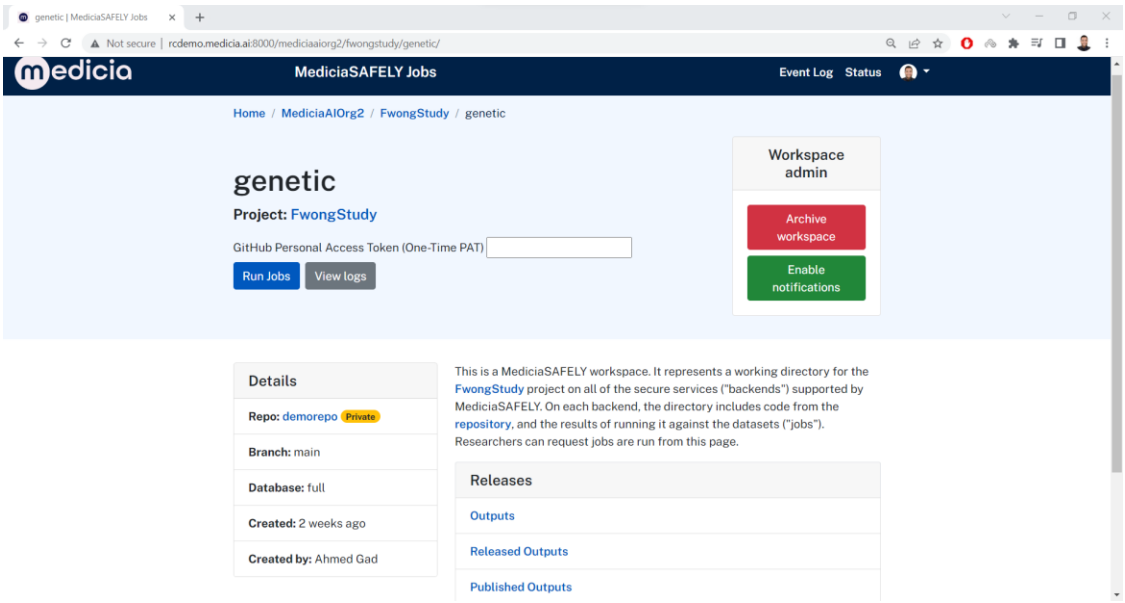


There are 4 dropdown menus that can be used to filter the actions. From left to right, the menus are:

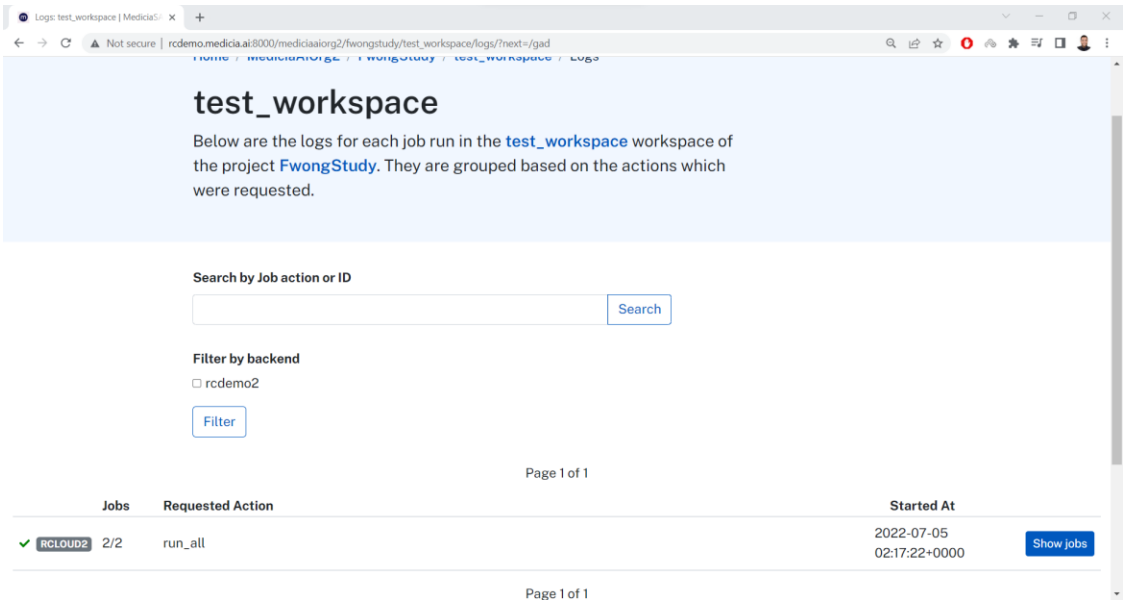
1. Backend: A list of all backends.
2. Job status: Can be either `failed`, `running`, `pending`, and `succeeded`.
3. User: A list of all users.
4. Workspace: A list of all workspaces.

Workspace Logs

The `Event Log` link shows the logs of all jobs across all workspaces. To only view the logs of a specific workspace, just visit the workspace page.



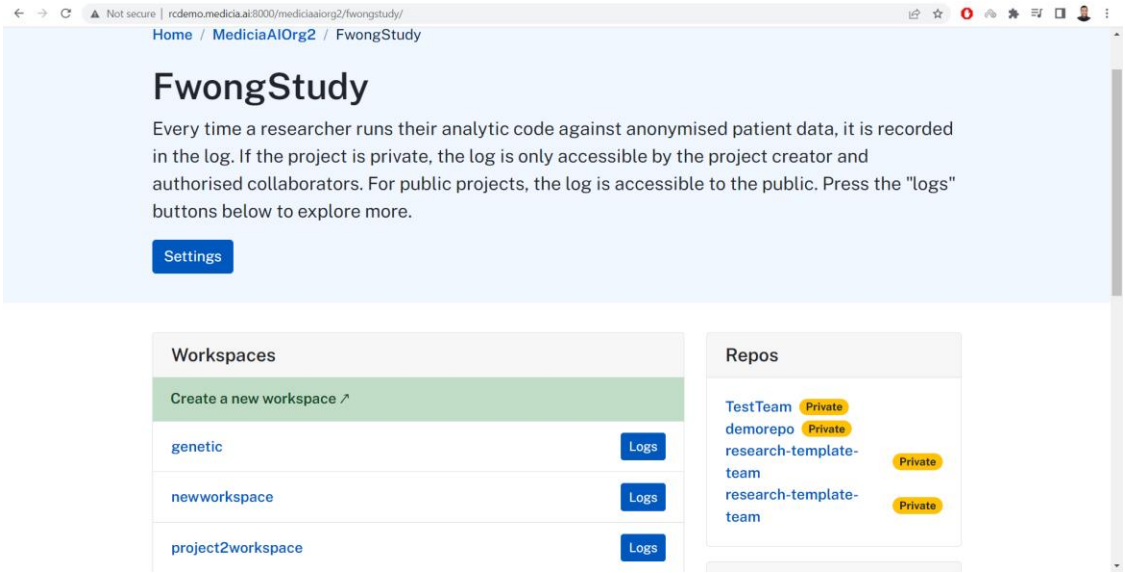
Below the workspace name, click the `View logs` button. It shows the logs for the current workspace only. Everything else is similar to what explained previously.



User Roles

The user can view the members of the project and all of their roles from the project page. For example, this is the page for the `FwongStudy` project.

Note that the `Repos` section shows the GitHub repositories associated with all the workspaces existing in this project.



Down in this page, there is a `Researchers` section where the members and their roles are listed.

FwongStudy | MediciaSAFELY Jo

Not secure | rcdemo.medicia.ai:8000/mediciaaiorg2/fwongstudy/

project2workspace Logs

test_workspace Logs

testworkspace Archived Logs

Created

2 months, 2 weeks ago

Organisation

MediciaAIOrg2

Researchers

Ahmed Gad

Governance Reviewer
Core Developer
Project Developer
Output Publisher
Onboarding Agent
Project Collaborator
Data Investigator
Project Coordinator
Technical Reviewer
Output Checker

fwong07

Project Coordinator